$y'' + (1+\sin^2 x)y' + (\cos^2 x)y = 3e^x$, $x \in [0;1]$, $y_{exact}(x) = e^x$

$\underline{x=0}$: $y(0) + y'(0) = 2$, $\underline{x=1}$: $1 \cdot y(1) + 0 \cdot y'(1) = e$

$h=1$, $x_0=0$, $x_1=1$: **Прямой ход**

$x=0$; $F(0)=3$, $q(0)=1$, $r(0)=1$, $\alpha(0)=-1$, $\beta(0)=2$ / $x=0,5$; $\alpha=-1,5$, $\beta=3,5$, $q\approx 1,23$, $r\approx 0,77$, $F\approx 4,95$

$K_{1,\alpha} = -(-1)^2 - 1(-1) - 1 = -1$

$K_{1,\beta} = -(-1+1)\cdot 2 + 3 = 3$

$K_{2,\alpha} = -(-1,5)^2 - 1,23(-1,5) - 977 = -1,1754$

$K_{2,\beta} = -(-1,5+1,23)\cdot 3,5 + 4,95 = 5,89$

$K_{3,\alpha} \approx -1,33829$

$K_{3,\beta} \approx 0,716$

$K_{4,\alpha} \approx -1,766$

$K_{4,\beta} \approx 13,648$

$\underline{x=1}$: $\alpha(1) \approx -1 + \frac{1}{6}(-1 + 2(-1,175) + 2(-1,338) - 1,766) = -2,299$

$\beta(1) \approx 2 + \frac{1}{6}(3 + 2(5,812) + 2(6,716) + 13,648) = 8,977$

Обратный ход: $h=1$, $y(1)=e$

$K_1 = \alpha(1)\cdot y(1) + \beta(1) \approx (-2,299)\cdot 2,718 + 8,977 = 2,728$

$y_{K_2} \approx 2,718 - \frac{1}{2} 2,728 = 1,354$; $K_2 = \alpha(0,5)\cdot y_{K_2} + \beta(0,5) \approx 2,122$

$y_{K_3} \approx 1,657$; $K_3 \approx 1,636$; $y_{K_4} \approx 1,082$; $K_4 \approx 0,918$

$y_n(0) \approx 2,718 - \frac{1}{6}(2,728 + 2(2,122) + 2(1,636) + 0,918) = 0,858$

$e_n = |y_n(0) - y_{exact}(0)| \approx |0,858 - 1| = 0,142$